Side A: the row `ic_online_indexbuild_r770`
source bc6a31c7… (PR #1006's frozen digest, re-derived)
state loop: prepare, 1 inversion per row, finish: 11 products/state
insert: lift_scan key of root $i{+}32$, prefetch, insert root i
self-test: full Feistel bijection check, 2 keys, 8 lanes

→

Side B: round 2 `ic_online_indexbuild2_r770`
source 4008a47a… = bc6a31c7… + 7 counted edits (invertible)
C9: $q = p\,(1/(\ell{+}r))^2$, $d = c\,(1/\ell + 1/r)^2$: 7 products/state
C7: linear run finder keys, 8 roots at a time, 32–40 ahead, prefetched
self-test: unchanged (the same full check)

Same output, gated per process and per base (K600, M38, M10): stored_states_blake3, table_slots_blake3, logical-state and table-map digests, root entries; the same relation, probes and scalar per point; the same rank tables; per rank seed the same relation digest, probe vector, root calls and counters (kM38IB, kM38IC, kM38ICaa).

Segment Pi (point s$(i{-}1)$ of 12, role indexbuild2-cold-v1); sub-block order shuffled by `random.Random(2026102070+`$i{-}1$`)`

primary: 3 pairs, strict alternation ABABAB or BABABA (seeded start; 8 points start A, 4 start B)



A/A: 2 pairs, alternating, seeds of primary pairs 1, 2

planted: 1 cold pair (lift_plant)

rho



Run order: O1 (Q0 on cPIB, cPIC, cM10IB, cM10IC, cM38IB, cM38IC: both producers on all three bases; exactness only) then P1 … P12; 282 fresh processes, each pinned to one CPU of one ubuntu-24.04 runner; time guard at 165 min, job limit 180 min (any skip: invalid). Side = rank process (rank stage with the pair's seed, then the point) followed by cold process (verified table, then the point). Same rank seed on both sides of a pair, not a shared rank stage.

Primary (frozen): $E2E(\text{side}) = \text{rank_stage}(k) + \text{cold}(c)$, $R = \dfrac{\sum_{36\ \text{pairs}} E2E(A)}{\sum_{36\ \text{pairs}} E2E(B)}$; 95% two-stage paired bootstrap (points, then pairs within points; A and B together), 10,000 draws, seed 20261040. **Improvement iff the lower bound > 1.** Stage secondary (labelled): index build cM38IB / cM38IC. Controls: A/A $E2E(B)/E2E(Baa)$ and cold inside $[0.97, 1.03]$; planted index build cM38ICX/cM38IC, lower bound > 1; planted closure corruptions rejected; planted non-bijections rejected (parent's crate tests). Every gate true (unknown = fail).